

NHS Digital Hospital Agent — Dataset Guide

Which datasets to download and use, and how to access them legally, for a UK NHS hospital AI agent project

Golden rule for the team: default to synthetic/open data for development. Real patient data should only enter the project once governance (DPIA, Data Sharing Agreement, Information Governance approval) is actually in place — not "in progress."

Tier 1 — Synthetic Data (use first, no approval needed)

Dataset	What it is	Where
Synthea	Open-source synthetic patient generator (demographics, encounters, conditions, meds) — configurable to mimic NHS-like populations	github.com/synthetichealth/synthea
NHS England Synthetic Data sets	NHS-published synthetic datasets designed to mirror real NHS data structures without real patient risk	NHS England Digital synthetic data releases (check digital.nhs.uk)
CPRD synthetic/dummy datasets	Structured like Clinical Practice Research Datalink extracts, for tooling development	cprd.com

Tier 2 — Open / Aggregate UK Health Data (no patient identifiers)

Dataset	What it is	Where
Hospital Episode Statistics (HES) — aggregate/open extracts	Admissions, A&E, outpatient activity at aggregate level	NHS England Digital open data portal
NHS England Open Data Portal	Performance stats, waiting times, workforce, activity data	opendata.nhs.uk / digital.nhs.uk
OpenPrescribing	Open GP prescribing data across England	openprescribing.net
ONS Health Data	Population, mortality, health inequalities statistics	ons.gov.uk
SNOMED CT UK Edition	The clinical terminology standard your agent should map to — free for NHS-affiliated use via TRUD licence	isd.digital.nhs.uk (TRUD)
NHS Data Dictionary	Official field/data-standard definitions for NHS systems — essential for schema design	datadictionary.nhs.uk
NHS England Data Uses Register	See what other organisations were approved to use similar data for — useful precedent	digital.nhs.uk

Tier 3 — Real / Pseudonymised NHS Data (requires formal approval)

Accessed via **DARS — the Data Access Request Service**, the official gateway for NHS health and social care data. Start this early: approval can take weeks to months.

- 1 **Enquiry** — contact NHS Digital/NHS England enquiries (enquiries@nhsdigital.nhs.uk / 0300 303 5678) before applying. Many needs are actually met by Tier 1/2 data instead.
- 2 **Application** — complete the DARS application with purpose, legal basis, and data-minimisation justification.
- 3 **Approval** — reviewed by a Case Officer / independent oversight panel.
- 4 **Access** — most access now happens through an **NHS England Secure Data Environment (SDE)**, where data never leaves the controlled environment (raw extracts are now the exception). Related services: **OpenSAFELY** (pseudonymised GP/NHS data analysis), **NHS DigiTrials** (clinical trials).
- 5 **Audit & Deletion** — ongoing audit obligations and data deletion at the end of the agreement.

Full guidance: digital.nhs.uk/services/data-access-request-service-dars

Tier 4 — General ML/Reference Datasets (baseline capability, not NHS-specific)

Dataset	What it is
MIMIC-IV	US ICU data, de-identified, free credentialed access via PhysioNet — good for general clinical NLP prototyping
i2b2 / n2c2 clinical NLP datasets	De-identified clinical notes, for entity-extraction tasks
HuggingFace Datasets — medical/clinical collections	Search "clinical", "medical-QA" — verify licence per dataset before use

Note: reflects standard NHS Digital / NHS England process and publicly available data-access routes as of August 2026. Processes and portal names occasionally change — verify current details on digital.nhs.uk before formal submissions, and treat this as a starting reference, not a substitute for sign-off from your organisation's Information Governance and Clinical Safety teams.